

Student Submission - Good Answer (Student ID: A001)

Answer to Question 1

Encapsulation is the OOP principle of bundling data (attributes) and the methods that operate on that data within a single unit called a class, while restricting direct access to some of the class's components.

Access modifiers such as private, protected, and public implement encapsulation. Private members can only be accessed within the class, protected members are accessible to subclasses, and public members are accessible from anywhere.

Encapsulation improves code maintainability because it hides the internal implementation details from the outside world. Changes to the internal implementation do not affect other parts of the code as long as the public interface remains unchanged. This reduces coupling and makes debugging easier.

Answer to Question 2

A class is a blueprint or template that defines the attributes and behaviors (methods) that its objects will have. It is a user-defined data type that encapsulates data and functions.

An object is an instance of a class. When a class is defined, no memory is allocated. When an object is created from the class, memory is allocated for the object's attributes.

The relationship between class and object: a class defines the structure, and objects are actual realizations of that structure. Multiple objects can be created from the same class, each with their own state.

Example: Consider a class Car with attributes color and speed, and a method drive(). An object myCar = new Car() is an instance of Car. Another object yourCar = new Car() is a different instance but follows the same blueprint.

Answer to Question 3

Inheritance is a mechanism in OOP where a new class (child class or subclass) derives attributes and methods from an existing class (parent class or superclass). The child class inherits all public and protected members of the parent class.

The parent-child relationship: the child class is a specialized version of the parent class. For example, class Dog extends Animal means Dog inherits from Animal.

One benefit of inheritance is code reuse. Instead of rewriting common

attributes and methods in every class, you define them once in the parent class and all child classes automatically have access to them. This reduces code duplication and makes the system easier to maintain.